

SIMULATION OF SELF-DRIVING CAR USING DEEP LEARNING



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CERTIFICATE OF COMPLETION

This is to certify that the research entitled “**Simulation Of Self-Driving Car Using Deep Learning**” submitted by **B. Sanjay, M. Chinna Rao** and **Sai Kishore** to the Department of Computer Science And Engineering, Rajiv Gandhi University Of Knowledge Technologies – Srikakulam Campus for the award of the degree of **B. Tech in CSE** is a Bonafide record of work carried out by them under my supervision. The contents of this research, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Project Guide

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Abstract — According to the WHO, road traffic crashes claim around 1.35 million lives annually. The advancements in artificial intelligence and its integration into the automotive industry have the potential to significantly reduce these numbers. In this project, we present the implementation of a self-driving car system using deep learning techniques. The car is simulated in the Udacity simulator to ensure safety and controlled testing. The dataset for training is generated within the simulator, pre-processed, and augmented to improve the model's performance. A Convolutional Neural Network (CNN) inspired by NVIDIA's architecture is developed to predict steering angles based on real-time camera input. The results demonstrate the model's ability to navigate diverse road conditions, achieving stable and accurate performance in the simulated environment.

Keywords: autonomous driving, deep learning, convolutional neural networks, data augmentation, self-driving car simulation, Udacity simulator.

I. Introduction

In the modern era, advancements in automation and artificial intelligence (AI) have significantly transformed various industries, offering innovative solutions to complex real-world problems. One of the most compelling applications of AI is in the development of autonomous driving systems, which have the potential to reduce road accidents, optimize traffic flow, and save valuable time for users.

Autonomous vehicles rely on a combination of machine learning algorithms and computer vision techniques to interpret their surroundings and make driving decisions. This project aims to design and implement a self-driving car model using convolutional neural networks (CNNs), a powerful deep learning approach for image recognition and feature extraction.

The project utilizes the Udacity simulator, an advanced tool that provides a safe and controlled environment for training and testing self-driving car models. The simulator allows the collection of driving data in the form of images from three perspectives: left, center, and right cameras. These images are accompanied by corresponding steering angle labels, enabling the model to learn the relationship between visual inputs and driving actions.

The CNN-based model is trained to predict steering angles within a range of -1 to 1, where negative values indicate left turns, zero represents no steering adjustment, and positive values signify right turns. The training process involves data augmentation and preprocessing to improve

model performance and robustness. Once trained, the model is tested in the simulator's autonomous mode to evaluate its ability to navigate through diverse tracks and conditions.



Figure 1 Main Screen of Simulator

This paper details the development process, the architectural design of the model, and its performance in the Udacity simulator. The findings demonstrate the potential of deep learning techniques in creating efficient and reliable autonomous driving systems.

II. Literature Review

In recent years, autonomous driving has been a significant area of research, leveraging advancements in artificial intelligence and deep learning. This section provides an overview of related works that have contributed to the field, focusing on self-driving car systems and convolutional neural networks (CNNs).

Shuyang Du et al. [1] explored the use of deep learning to predict steering angles for self-driving vehicles, emphasizing image recognition techniques such as convolutional neural networks (CNNs). Their work demonstrated the effectiveness of using CNNs for steering angle prediction, using images captured from the vehicle's front-facing camera during driving. This work is essential for understanding how image-based approaches can drive vehicle control in autonomous driving systems.

Bojarski et al. [2] introduced an end-to-end deep learning approach for self-driving cars, where a CNN maps raw input images directly to steering commands. Their study used minimal training data and demonstrated that the system could drive on local roads, highways, and parking lots, even in areas with unclear visual guidance. This approach highlighted the power of CNNs in self-driving tasks, paving the way for future research in end-to-end learning for autonomous driving.

The research by Yang et al. [3] expanded the scope of autonomous vehicle control by incorporating multi-modal and multi-task approaches. They utilized visual perceptions to control the vehicle, emphasizing the importance of handling various tasks simultaneously, such as steering, speed regulation, and obstacle avoidance. Their work showcases the complexity of autonomous driving tasks and the need for multi-task learning to ensure robustness in real-world driving environments.

Rafael de Jesus Guedes [4] explored the combination of computer vision and explainable reinforcement learning (RL) for autonomous driving. By integrating RL with computer vision, their system was able to learn driving behaviors through interactions with the environment, while also providing explanations for decision-making, a critical aspect of improving trust and safety in autonomous systems. This approach represents an important step towards making autonomous driving systems more interpretable.

Pérez-Gil et al. [5] investigated the application of deep reinforcement learning (DRL) in controlling autonomous vehicles in the CARLA simulator. They demonstrated how DRL techniques could be applied to improve the control of vehicles in complex simulated environments, highlighting the potential for reinforcement learning to address challenges in vehicle control, navigation, and decision-making.

Finally, Ahire et al. [6] explored the use of the CARLA simulator for vehicle driving simulation. By leveraging this platform, they could test and simulate vehicle behavior in various driving conditions, providing valuable insights into the development of autonomous driving systems. Their work illustrates the importance of realistic simulation environments in training and evaluating autonomous driving models.

These studies collectively contribute to the ongoing development of self-driving vehicles by advancing techniques in image recognition, deep learning, reinforcement learning, and simulation-based testing. Each work highlights different facets of the problem, offering solutions that range from image-based steering prediction to complex, multi-modal control systems for autonomous driving.

III. Methodology

The development of the self-driving car system in this project involves several critical stages. Each stage is designed to systematically collect, process, train, and evaluate the data for effective implementation. Below are the key steps of the methodology, explained in detail:

A. Data Collection

For data collection, we used Udacity's self-driving car simulator, a powerful tool designed to provide a virtual environment for training and testing autonomous driving systems. The simulator includes two modes: training mode

and autonomous mode. In the training mode, we manually drove the car on Track 1, which is one of the predefined tracks.



During this process, the simulator captured images from three cameras mounted on the car: the centre, left, and right. This technique, known as behavioural cloning, involves replicating the user's driving behaviour based on the recorded data.



Figure 2 Images captured from left, center and right side of the car

The images from the three cameras are critical for training the model to recover from positional drifts. For instance,

[illegible][illegible]

B. Data Preprocessing

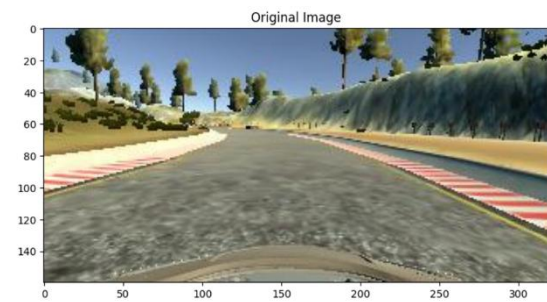
The histogram shows the frequency distribution of the variable 'score'. The x-axis represents the score values from -1.00 to 1.00, and the y-axis represents the frequency from 0 to 3000. The distribution is centered at 0.00, with a peak frequency of approximately 3000. A horizontal line is drawn at y = 400.

A histogram showing the frequency distribution of the variable 'score'. The x-axis represents the score values, ranging from -1.00 to 1.00 with major ticks every 0.25. The y-axis represents the frequency, ranging from 0 to 400 with major ticks every 50. The distribution is unimodal and slightly right-skewed, with the highest frequency (approximately 400) occurring at the score of 0.00. The bars are blue, and there is a horizontal blue line at the top of the y-axis at 400.

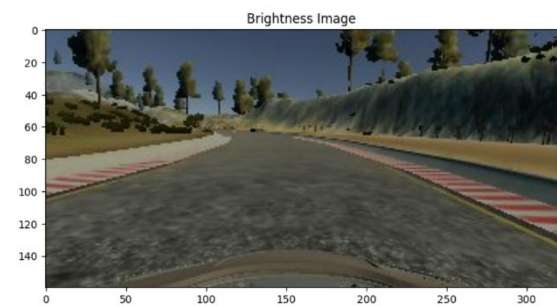
The next step involves converting the images to the YUV colour space, which is commonly used in computer vision tasks for self-driving cars due to its robustness under varying lighting conditions. The Y channel represents luminance, while the U and V channels represent chrominance. This conversion helps the model focus on the structural details of the road.

C. Data Augmentation

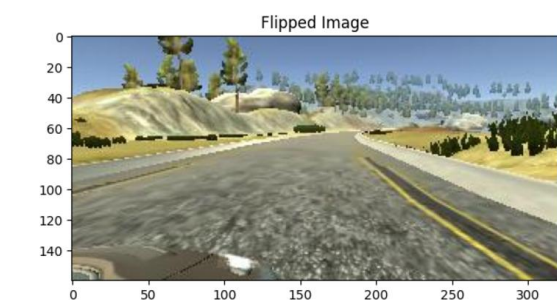
To improve the generalizability of the model, data augmentation techniques are applied to simulate a variety of driving scenarios. These techniques include:



Brightness Adjustment: Modifying the brightness of images to mimic different lighting conditions, such as day and night driving.

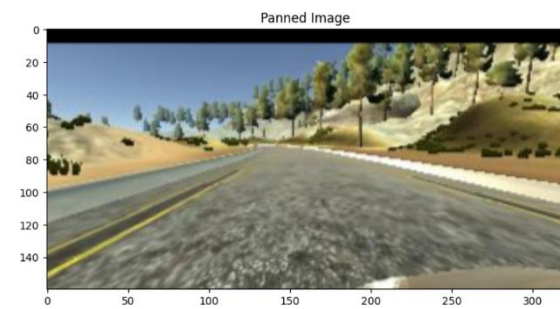
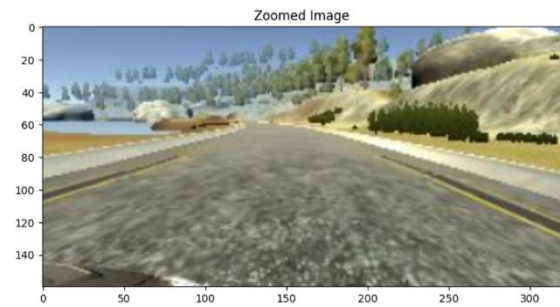


Flipping: Horizontally flipping the images and adjusting the steering angles accordingly to simulate driving in the opposite direction.



Shadow Addition: Adding synthetic shadows to mimic real-world road conditions under trees or bridges.

Zooming and Panning: Applying slight zoom-in or zoom-out effects and translating the image to simulate camera displacements.



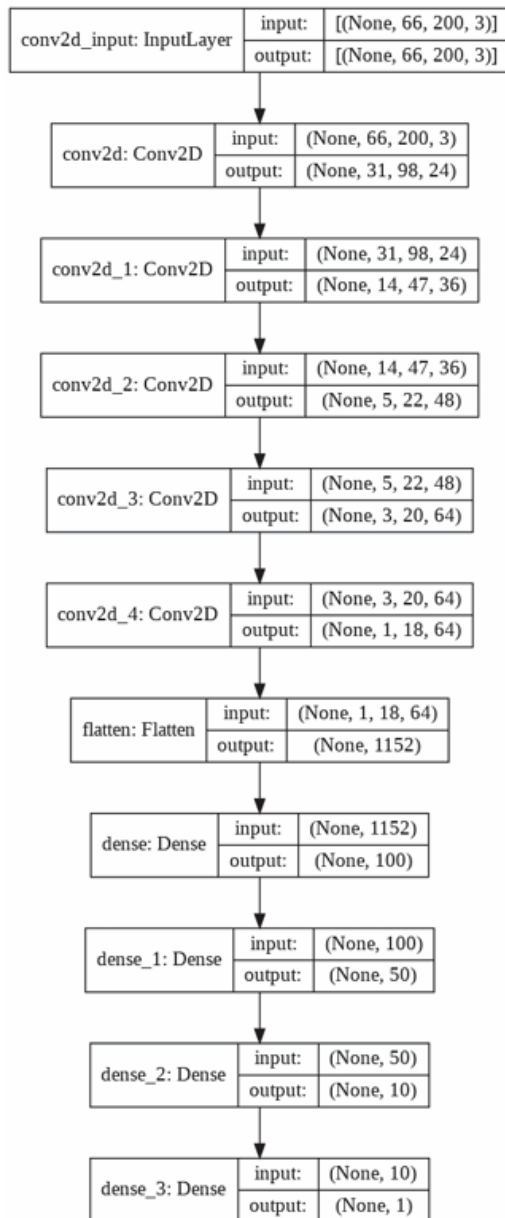
By augmenting the dataset, the model learns to handle diverse conditions, making it more robust and reliable.

D. Model Architecture & Training

The training process involves feeding the pre-processed and augmented dataset into a convolutional neural network (CNN). We used NVIDIA's CNN architecture, specifically designed for self-driving applications. The model consists of several convolutional layers for extracting spatial features from the images, followed by fully connected layers for predicting the steering angle.

The input layer receives images with dimensions 200x66x3 (height, width, and colour channels). Convolutional layers use filters to detect edges, curves, and other features in the images, while pooling layers down sample the feature maps to reduce computational load. Fully connected layers at the end of the network combine these features to output a single value: the steering angle.

Epoch 1/10	
300/300	132s 419ms/step - loss: 0.6057 - val_loss: 0.0990
Epoch 2/10	
300/300	127s 424ms/step - loss: 0.2730 - val_loss: 0.1155
Epoch 3/10	
300/300	125s 418ms/step - loss: 0.1185 - val_loss: 0.1160
Epoch 4/10	
300/300	124s 415ms/step - loss: 0.1153 - val_loss: 0.1189
Epoch 5/10	
300/300	126s 422ms/step - loss: 0.1130 - val_loss: 0.1162
Epoch 6/10	
300/300	117s 391ms/step - loss: 0.1155 - val_loss: 0.1164
Epoch 7/10	
300/300	118s 394ms/step - loss: 0.1165 - val_loss: 0.1156
Epoch 8/10	
300/300	118s 395ms/step - loss: 0.1168 - val_loss: 0.1173
Epoch 9/10	
300/300	118s 396ms/step - loss: 0.1136 - val_loss: 0.1145
Epoch 10/10	
300/300	118s 396ms/step - loss: 0.1122 - val_loss: 0.1139

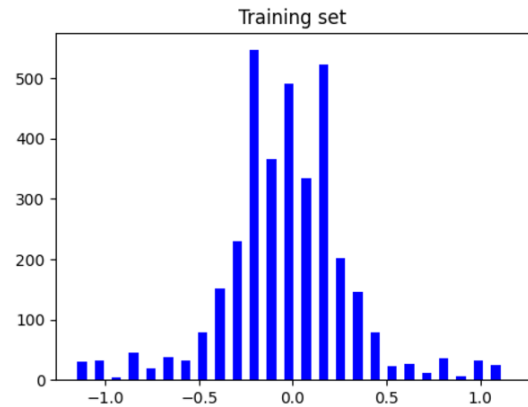


The model is trained by minimizing the **mean squared error (MSE)** between the predicted and actual steering angles. The Adam optimizer is employed for efficient gradient descent, and the learning rate is tuned to balance convergence speed and stability. The dataset is split into training and validation sets, with 80% used for training and 20% for validation.

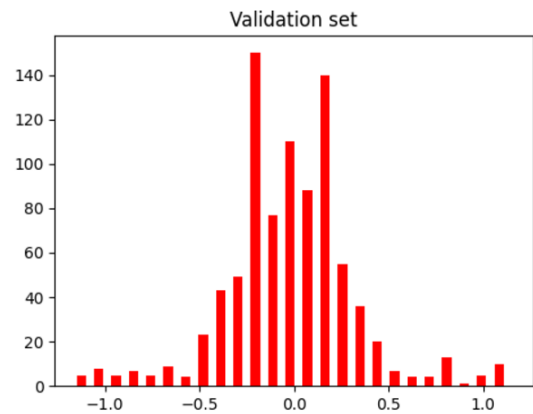
Layer (type)	Output Shape	Params #
conv2d_4 (Conv2D)	(None, 1, 18, 64)	1,280
conv2d_5 (Conv2D)	(None, 14, 47, 36)	11,536
conv2d_6 (Conv2D)	(None, 5, 22, 48)	6,240
conv2d_7 (Conv2D)	(None, 3, 20, 64)	2,720
Flatten_1 (Flatten)	(None, 1152)	0
dense_4 (Dense)	(None, 100)	116,200
dense_5 (Dense)	(None, 50)	5,050
dense_6 (Dense)	(None, 10)	510
dense_7 (Dense)	(None, 1)	11

E. Model Testing and Validation

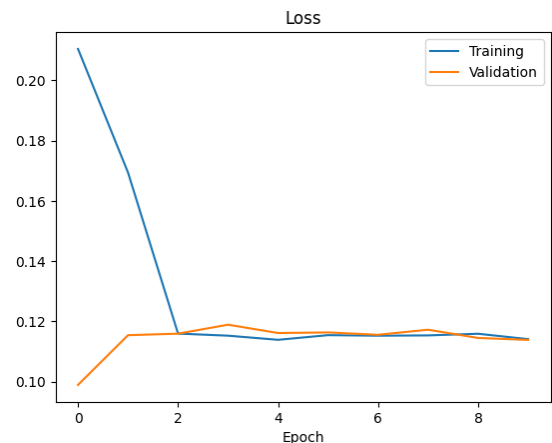
We have used 1000 images for training and 251 images for testing our model.



With the use of matplotlib library available in python, we have plotted a chart as shown in below figure. It visualizes the count of steering angles in the testing dataset.



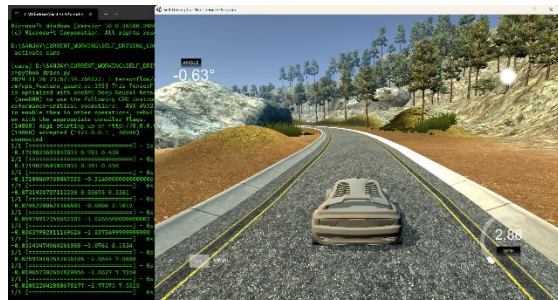
After training, the model's performance is validated by deploying it in the simulator's **autonomous mode**. In this mode, the model predicts steering angles based on real-time input images from the simulator. These predictions are used to control the car's steering, throttle, and braking systems.



The model is tested on both the training track and unseen tracks to evaluate its generalization capability. Metrics such as track completion rate, deviation from the lane centre, and recovery from positional drifts are used to assess the model's effectiveness.

F. Deployment

Finally, the trained model is deployed using a Python script that connects to the simulator via a WebSocket interface. The script preprocesses images from the simulator in real time, feeds them into the model, and sends the predicted steering angle and throttle commands back to the simulator. This setup ensures a smooth transition from development to testing in a simulated environment.



Through this structured methodology, we developed a robust and efficient self-driving car system capable of autonomous navigation in a simulated environment.

IV. Conclusion

In this project, we successfully implemented a self-driving car model using deep learning techniques. By collecting data from Udacity's simulator, we trained a convolutional neural network (CNN) to predict steering angles based on images captured from the car. The model was able to navigate predefined tracks autonomously with high accuracy.

Our approach demonstrated the potential of deep learning in autonomous driving, showcasing its ability to replicate human driving behaviour. While promising, further improvements can be made by using larger datasets and adding more sensor data for enhanced performance in real-world scenarios.

V. References

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